LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 11: From Blocks to Python

Module 4 | 45 minutes | Bridging to text code | Grades 6–8 · ages 12–14

Objective: Translate between Scratch blocks and Python: variables, loops, conditionals, lists and functions — and implement a classic algorithm from Python code.

Success criteria

- I can translate between Scratch and Python.
- I can implement an algorithm from Python code.
- I can name key Python syntax rules.

Key vocabulary

- **syntax**: The rules for writing code in a language.
- **indentation**: Spaces at the start of a line that group code.
- **elif**: Python's "else if".
- **zero-based**: Counting positions from 0.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Translation cards (5 min): match Scratch block cards to Python line cards.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	FizzBuzz from Python: learners follow the steps.
14 min	**Perform**	Match the Scratch block to its Python, Code Lab: FizzBuzz, Trace the Python
9 min	**Probe**	Python essentials, activity
4 min	**Wrap-up**	Exit ticket: Write repeat (10) in Python.

Unplugged warm-up

Translation cards (5 min): match Scratch block cards to Python line cards.

Teaching the idea

Explicit mapping of Scratch constructs to Python syntax: assignment, for-range, while, if/elif/else, def, lists (0-based), indentation as block structure. FizzBuzz exercises ordering of conditions and modulo.

Analogy: Translating a sentence: the meaning stays, the words change.

Live demo: Put a Scratch script and its Python side by side; colour-match each line.

Questions to ask

Ask	Listen for
What does indentation do in Python?	Shows which lines are inside.
What is range(1, 16)?	1 to 15.
Why check 15 first in FizzBuzz?	Otherwise 15 says Fizz.

Differentiation

Support
Provide the loop; learners write the nested if-else.

Stretch
Write the Python for your Search Showdown on paper.

Common misconceptions

Misconception	How to fix it
Python counts from 1	Lists and strings start at 0.
Indentation is just style	In Python it changes meaning.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write repeat (10) in Python.

Homework
Try a free online Python editor and type the FizzBuzz program.

Answer key — Kitabash activities

1. Match the Scratch block to its Python (perform · matching)

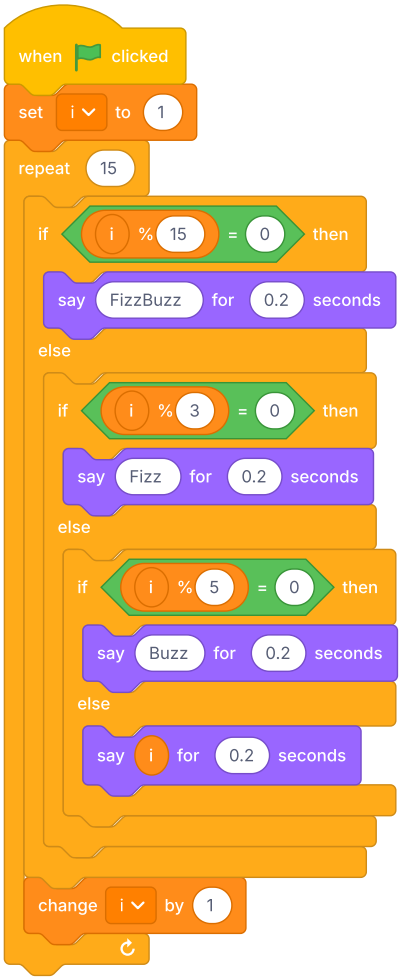
set [x v] to (5) → x = 5 · repeat (10) → for i in range(10): · forever → while True: · say [Hi] → print("Hi") · define greet (name) → def greet(name):

2. Code Lab: FizzBuzz (perform · code lab)

Goals checked: Correct sequence · Uses mod · Uses a loop

One solution:

Coder



3. Trace the Python (perform · code output)

4. Python essentials (probe · fill blank)

Answers: def · indentation · 0

5. Explain your thinking (probe · free response)

Look for: blocks, python, easier. **Example:** Blocks are easier for beginners because you can't make spelling or bracket mistakes. Python is easier for long programs because typing is faster than dragging, and you can copy and search code — for example writing a 100-line game.

Answer key — worksheet

1. **Key words:** syntax = The rules for writing code in a language. · indentation = Spaces at the start of a line that group code. · elif = Python's "else if". · zero-based = Counting positions from 0.
2. **Match the Scratch block to its Python:** set [x v] to (5) → x = 5 · repeat (10) → for i in range(10): · forever → while True: · say [Hi] → print("Hi") · define greet (name) → def greet(name):
3. **Plan your code:** Goals: Correct sequence; Uses mod; Uses a loop.
4. **Trace the Python:** A. 16
5. **Python essentials:** def · indentation · 0

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.